

Chapter 4 – Basic Stochastic Models

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1 Introduction

1. In this Chapter, we look at some of the basic stochastic models.
2. Stochastic means random, with the origins of the word coming from German and Greek.
3. We will look at white noise and do some simulations in this Chapter.
4. By doing some simulations, we know exactly what sort of data we are generating, unlike real data where we can only guess by looking at the time series plot visually and at the correlogram.
5. Since we know exactly what the model of our simulated data is, by looking at the plots and the correlogram, we can have a better idea how the correlogram of certain types of stochastic models look like.
6. Hence, when we use real data, we can compare the correlogram with the theoretical correlogram, and thus suggest a suitable model for the data.
7. Also when we fit models, including time series models, we want the residual errors between the observations of a model y_t and the fitted values $\hat{y}_t$ to be completely randomly normally distributed. So a plot of the time series of the residuals should look like a plot for white noise and the correlogram of the residuals should also look like white noise correlogram.

2 Gaussian White Noise

1. A residual error w_t of a time series model at time t is the difference between the model y_t and the fitted value of the model $\hat{y}_t$. We write

$$w_t = y_t - \hat{y}_t.$$

2. The residual errors form a time series $\{w_1, w_2, \dots, w_n\}$.
3. A time series $\{w_1, w_2, \dots, w_n\}$ is said to be *discrete Gaussian white noise* if the variables w_i are independently and identically distributed as normal $N(0, \sigma)$.
4. This implies that all of them have zero mean, variance σ^2 and $\text{Cov}(w_i, w_j) = 0$ for $i \neq j$.
5. **Example 1.** Let us do one simulation of a standard Gaussian white noise series of length 100. For standard Gaussian white noise, every w_i is distributed as normal $N(0, 1)$. You can type `help(rnorm)` to see how to set the parameters to simulate other types of normal distributions.

```
> set.seed(1)
> w <- rnorm(100)
> plot(w, type = "l", xlab="Time")
```

In my case, I obtained the plot as shown in Figure 1.

Now let us see whether this results look approximately normally distributed.

```
> x <- seq( -3,3, length = 1000)
> hist(w, prob = T); points(x, dnorm(x), type = "l")
```

The curve in Figure 2 is the ideal $N(0, 1)$ probability density curve. The histogram gives the distribution of our simulated results.

Figure 1: Plot of Gaussian White Noise

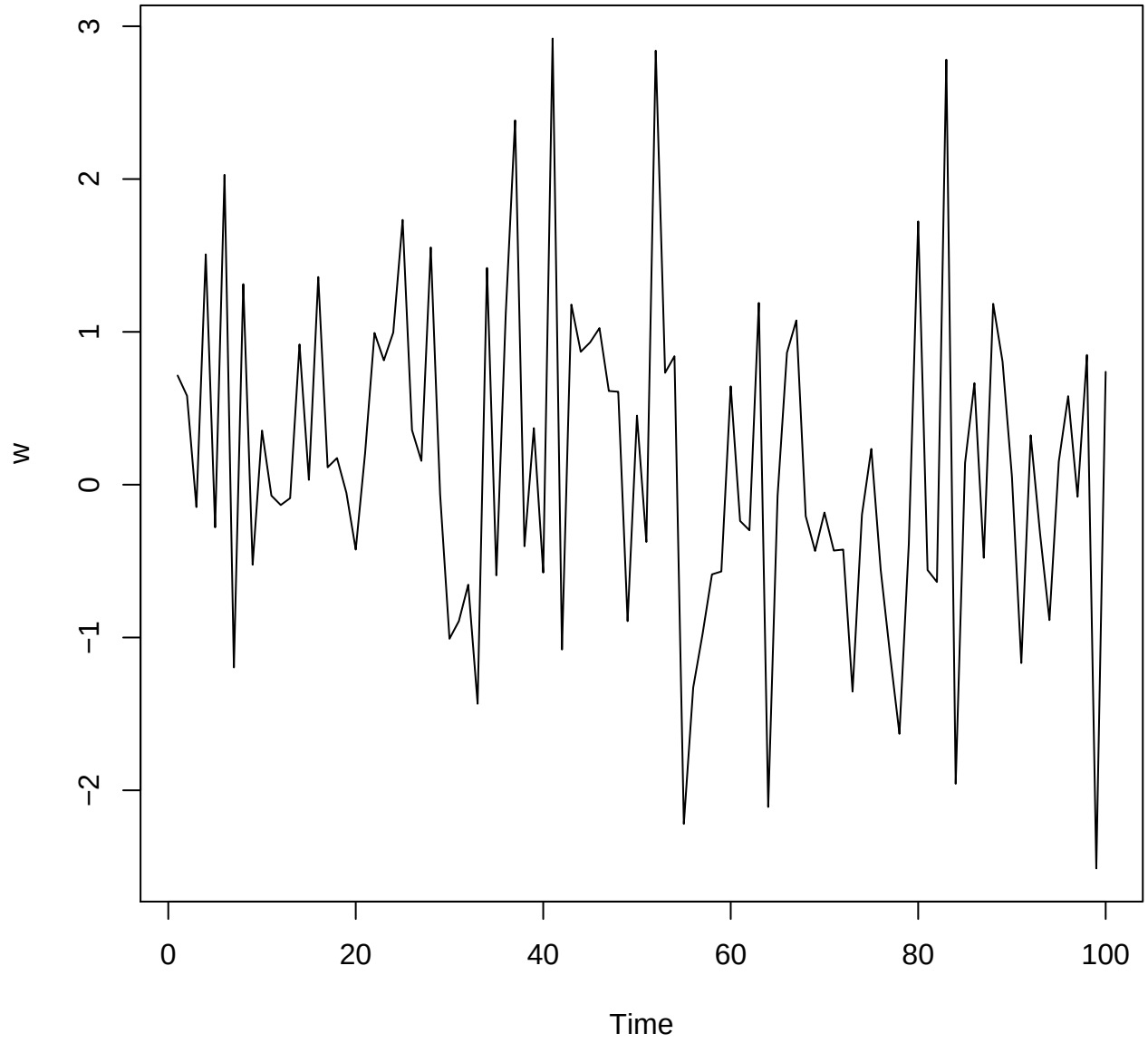
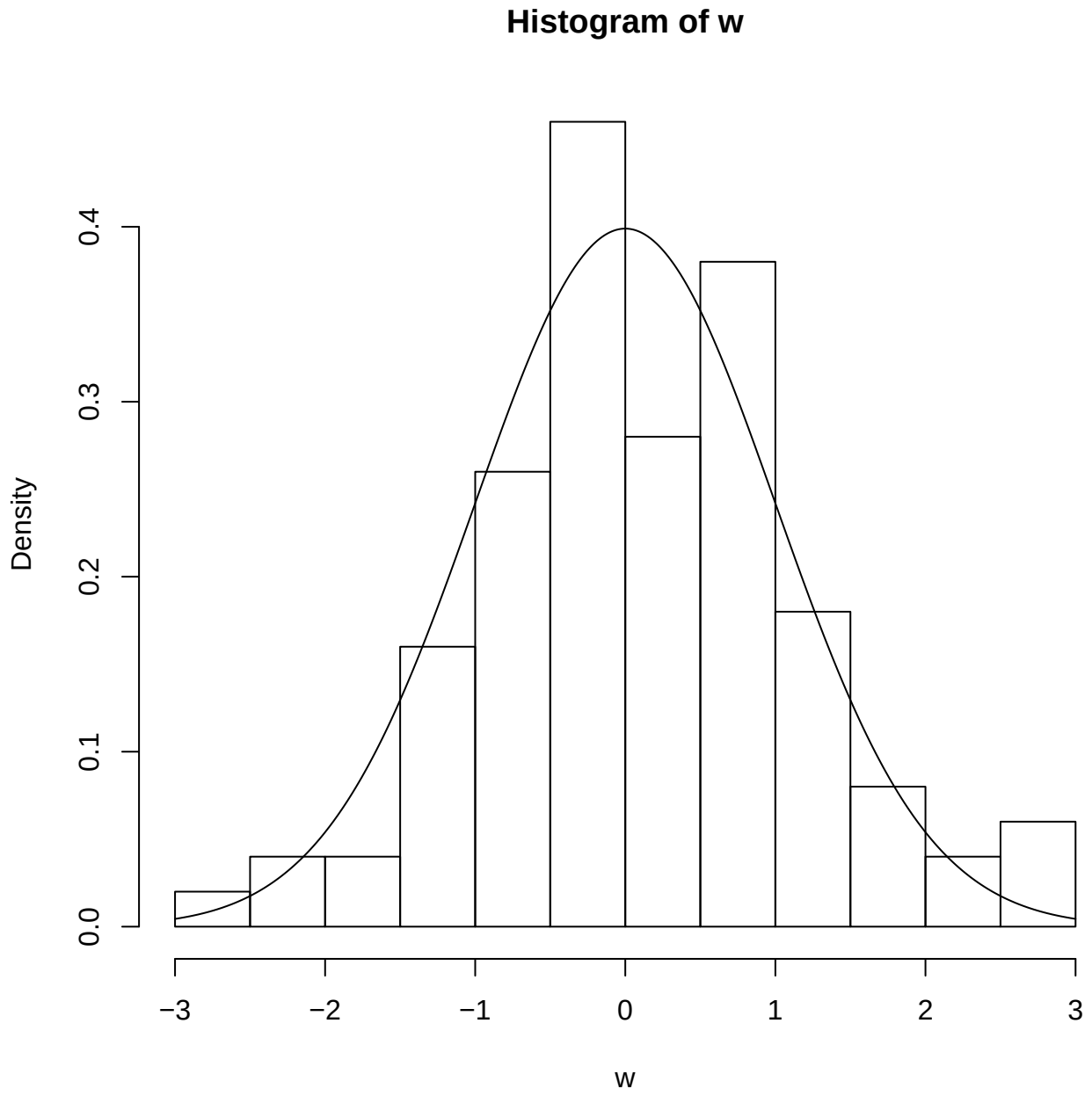


Figure 2: White Noise Histogram



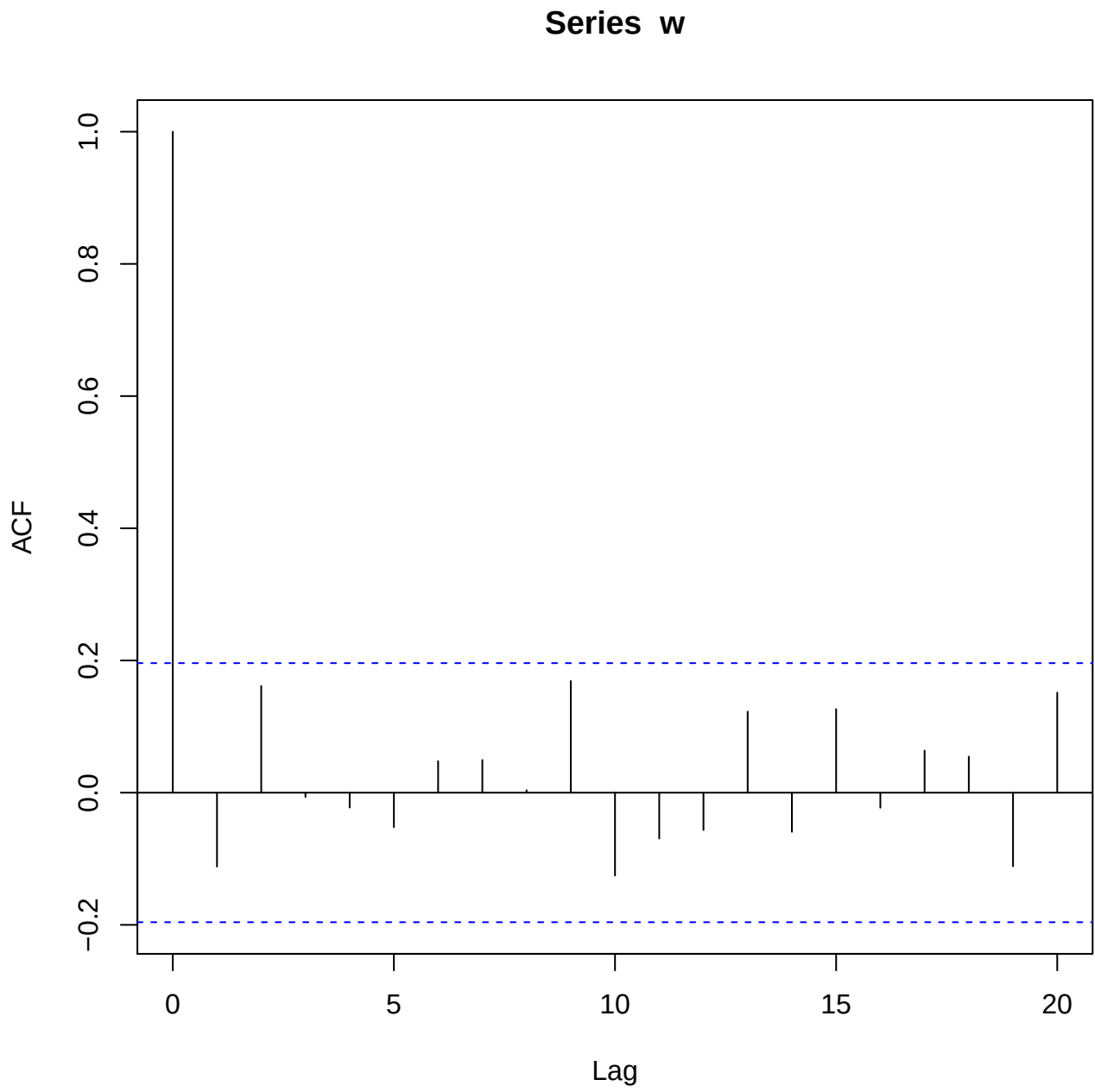
What I have obtained does not look quite normally distributed yet. It is possible that what you have simulated may actually look normally distributed. However, if you increase the simulation length to say, 1000, it will look more normal.

Let us see how the correlogram of our simulation looks like using the `acf(w)` command. You will get a plot that looks like the one in Figure 3.

In general, if for any time series data, you obtain a correlogram that looks like Figure 3 where it shows that an observation w_i is perfectly correlated to itself with $\rho_0 = 1$ (at lag 0) and all other correlations are insignificant (lying within the blue lines), then the time series is white noise.

6. After fitting a model, we can check that we have accounted for all trends and/or seasonality in the fitted model by looking at the residual errors series. What is left must be white noise for the residual errors and it is easy to use a correlogram of the residuals to check it. If need be, we can also run a Q-Q plot to check that it is normally distributed (see Chapter 2).

Figure 3: White Noise Correlogram



3 Random Walks

1. Let $\{x_t\}$ be a time series. Then $\{x_t\}$ is a *Gaussian random walk* if

$$x_t = x_{t-1} + w_t,$$

where $\{w_t\}$ is a Gaussian white noise time series.

2. From the previous item 3.1, we see that $x_{t-1} = x_{t-2} + w_{t-1}$ and so on, and so by successive backward substitution

$$x_t = w_1 + w_2 + \cdots + w_t.$$

3. If the initial starting value of x_0 is not 0, then also $x_t = x_0 + \sum_{i=1}^t w_i$. But without loss of generality, we can do our modelling by assuming that $x_0 = 0$ and adding on x_0 if necessary.
4. Here is how to simulate a Gaussian random walk using the stored simulation of the white noise $\{w_t\}$ earlier and the `cumsum()` command.

```
> plot(cumsum(w), type="l", xlab="Time")
```

The plot is shown in Figure 4.

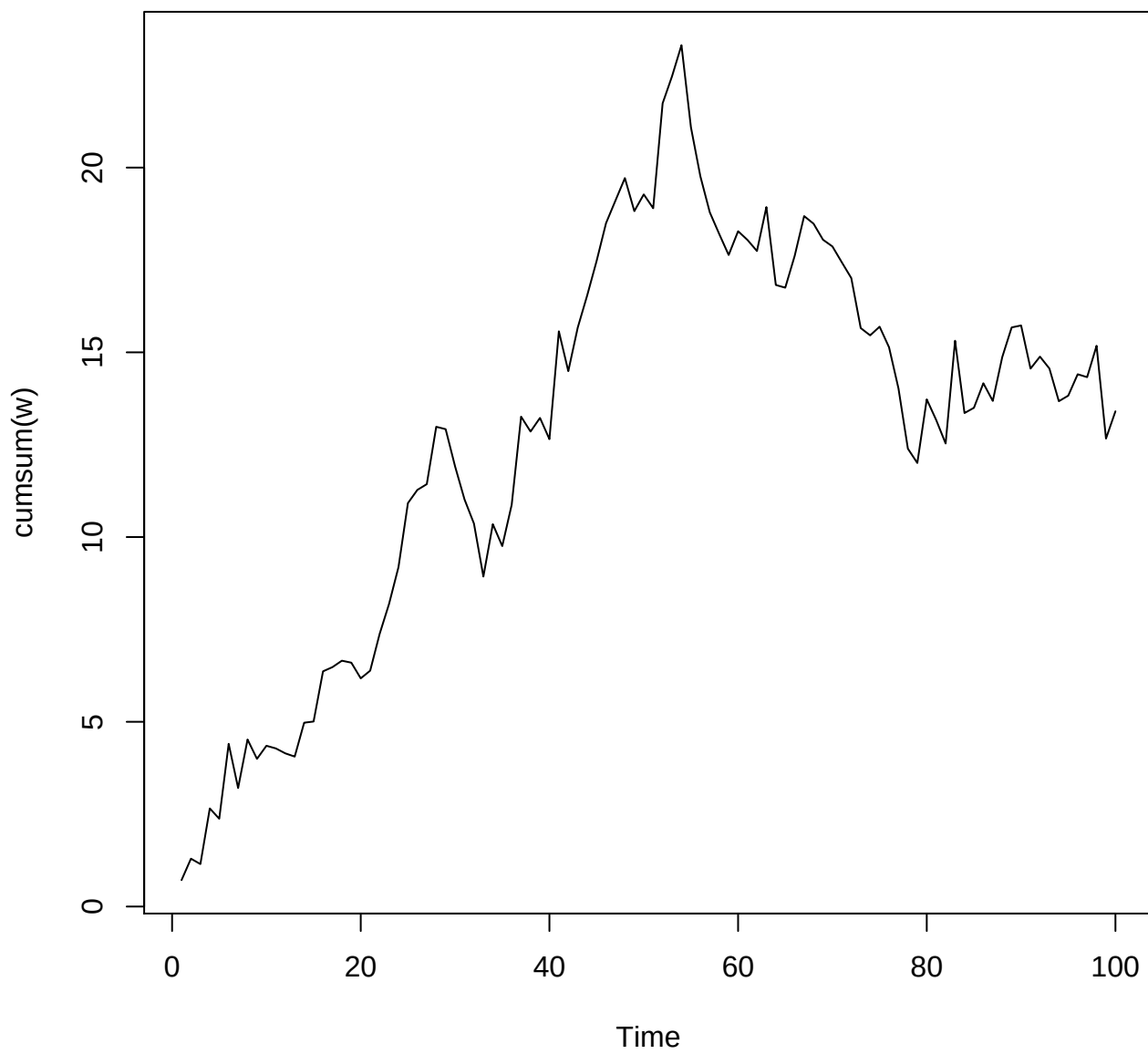
5. If you try doing a correlogram of your simulated random walk (`acf(cusmsum(w))`), you will see that it is decreasing linearly. The command `acf(cusmsum(w))[]` lists out all the autocorrelation values ρ_k up to lag 20 in this case.
6. Sometimes it is convenient to use the *backward shift* operator B . The backward shift operator B (also called the lag operator) shifts the observations back by one period as given by

$$Bx_t = x_{t-1}.$$

7. By repeatedly applying the operator B for n times, we get

$$B^n x_t = x_{t-n}.$$

Figure 4: A Simulated Random Walk



8. We can recover the equation in item 3.2 using the backward shift operator.

$$x_t = Bx_t + w_t \implies (1 - B)x_t = w_t \implies x_t = (1 - B)^{-1}w_t,$$

treating B like any regular algebraic variable. Since

$$(1 - B)^{-1} = 1 + B + B^2 + \cdots,$$

then we get

$$x_t = (1 + B + B^2 + \cdots)w_t = w_t + w_{t-1} + w_{t-2} + \cdots,$$

and since $w_0 = 0$ and $w_k = 0$ for negative k , then

$$x_t = w_t + w_{t-1} + w_{t-2} + \cdots + w_2 + w_1$$

as in item 3.2.

9. Now suppose we have the random walk x_t , we can define the *difference operator* ∇ (pronounced as “nabla”) by

$$\nabla x_t = x_t - x_{t-1}.$$

10. Since $x_t - x_{t-1} = w_t$ for $t = 2, 3, \dots$, then $\nabla x_t = w_t$ for $t = 2, 3, \dots$, and $\nabla x_1 = x_1 - x_0 = x_1$ if $x_0 = 0$.
11. Also using the backward shift operator, $\nabla x_t = x_t - x_{t-1} = (1 - B)x_t$ and for higher order differencing,

$$\nabla^n = (1 - B)^n = \sum_{k=0}^n \binom{n}{k} (-B)^k.$$

12. In **R**, using the `diff()` function gives us the different operator of order 1 by default. Thus

```
> diff(cumsum(w))
```

gives us the 1st order difference of the random walk $\nabla x_t = w_t$ for the values of w_t that we have obtained earlier in Example 1 for $t = 2, 3, \dots, 100$. You can compare the output with the output of w .

13. For higher order differences, e.g., 2nd order, this is the difference of the difference, i.e.,

$$\nabla x_t - \nabla x_{t-1} = x_t - x_{t-1} - (x_{t-1} - x_{t-2}) = x_t - 2x_{t-1} + x_{t-2},$$

which is exactly $\nabla^2 x_t$ as defined in item 3.11.

4 Random Walks with Drift

1. When there is drift, the relationship between the terms x_t and x_{t-1} in a random walk with drift is given by

$$x_t = x_{t-1} + \delta + w_t,$$

where $\{w_t\}$ is a Gaussian white noise process and $\delta \neq 0$ is the constant one-period drift term.

2. If $\delta > 0$, then the drift is positive, otherwise it is negative.
3. From item 4.1, we see that $x_{t-1} = x_{t-2} + \delta + w_{t-1}$, so by substitution back into 4.1, we obtain

$$x_t = x_{t-2} + 2\delta + w_t + w_{t-1},$$

and by subsequent backward substitution, we obtain

$$x_t = t\delta + \sum_{i=1}^t w_i.$$

This last equation implies that on average the process is drifting linearly upwards or downwards (depending on the sign of δ) since the mean of $\sum_{i=1}^t w_i$ is zero, however its spread as it proceeds forward in time also increases since the variance of $\sum_{i=1}^t w_i$ is proportional to t .

4. **Example 2.** Hewlett-Packard Company stock price data. The data consists of closing prices in USD for 672 trading days up to June 7, 2007. Note that for stock prices, in going from a Friday closing price to a Monday closing price, that is considered a 1 day time lapse since no trading occurs during the weekends and so we regard it as if time has been “suspended”. Likewise if there is a public holiday. Thus when dealing with stock price data, 1 trading year is around 252 days. We read and attach the data.

```
> HP.dat <- read.table("D:/MATH 5045/Data/HP.txt", header = T)
> attach(HP.dat)
```

We plot the stock price time series as shown in Figure 5.

```
> plot(ts(Price), xlab="Day", ylab="Price")
```

We can see that there appears to be a linearly upward trend.

So now let us look at its 1st order (or lag 1) difference using the `diff()` function and its plot in Figure 6. We see that the trend has more or less been removed.

```
> DPrice <- diff(Price)
> plot(ts(DPrice), xlab="Day", ylab="Difference of Closing Price")
```

By checking its correlogram with the `acf(DPrice)` command, we get the correlogram of the difference in prices in Figure 7. We see in Figure 7 that all the autocorrelations are more or less insignificant. The one corresponding to lag 10 appears to be just marginally significant.

In order to estimate the single day drift δ , this is the mean of `DPrice` and a 95% confidence interval for `DPrice` can also be calculated.

```
> mean(DPrice)
[1] 0.03986587
> mean(DPrice)+c(-1.96,1.96)*sd(DPrice)/sqrt(length(DPrice))
[1] 0.005088027 0.074643717
```

Figure 5: Hewlett-Packard Stock Price

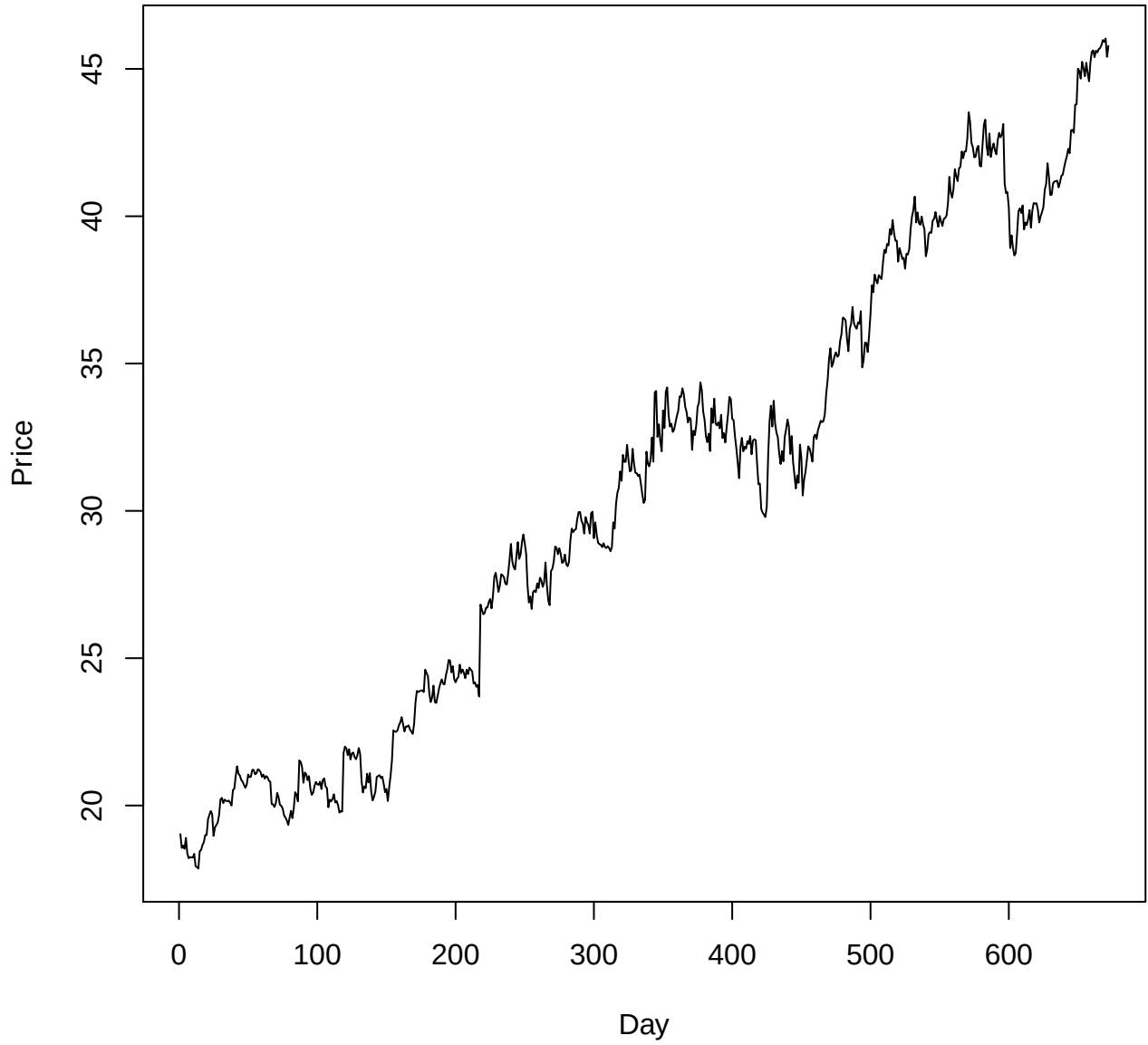


Figure 6: Lag 1 Difference of Hewlett-Packard Stock Prices

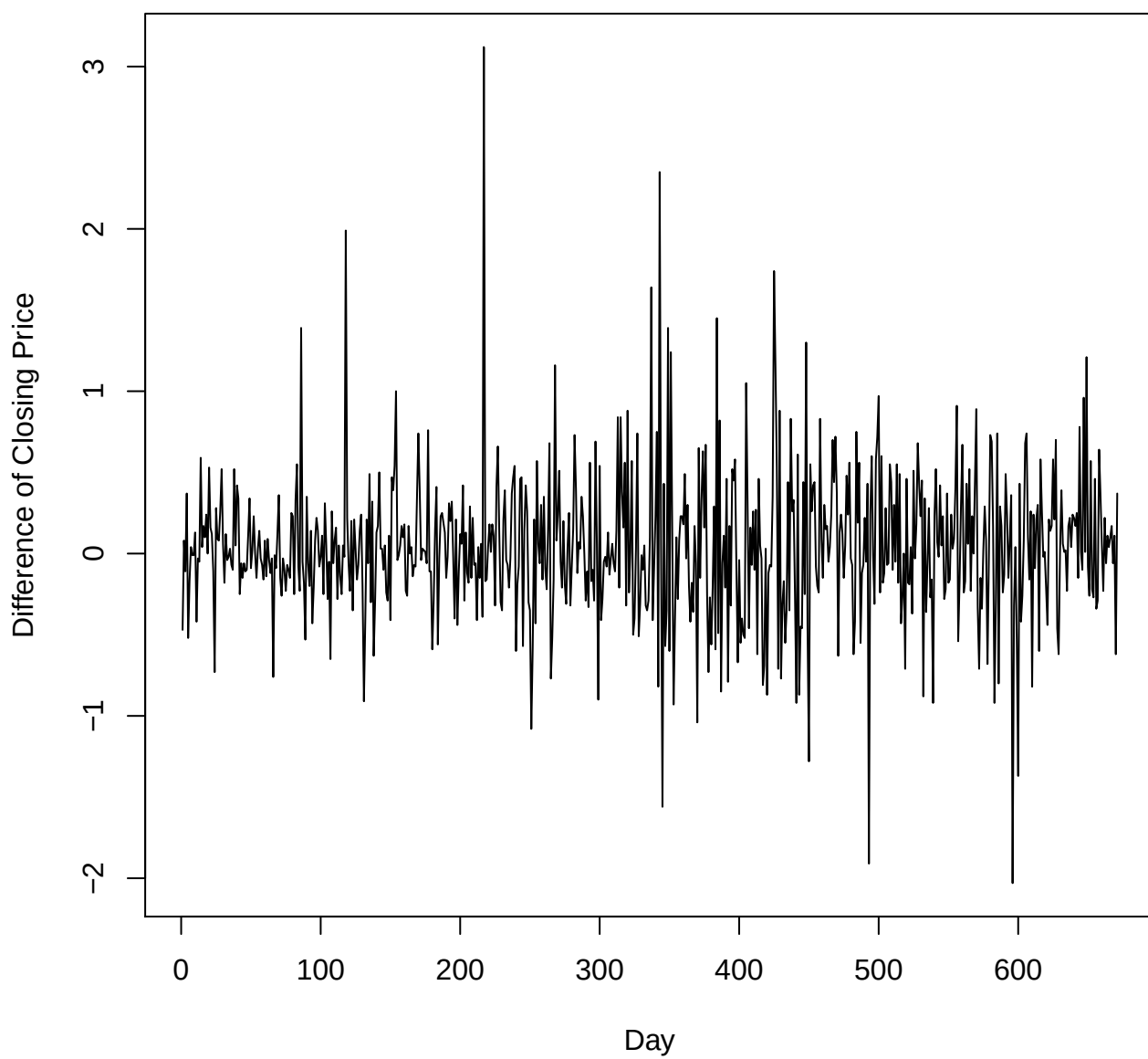
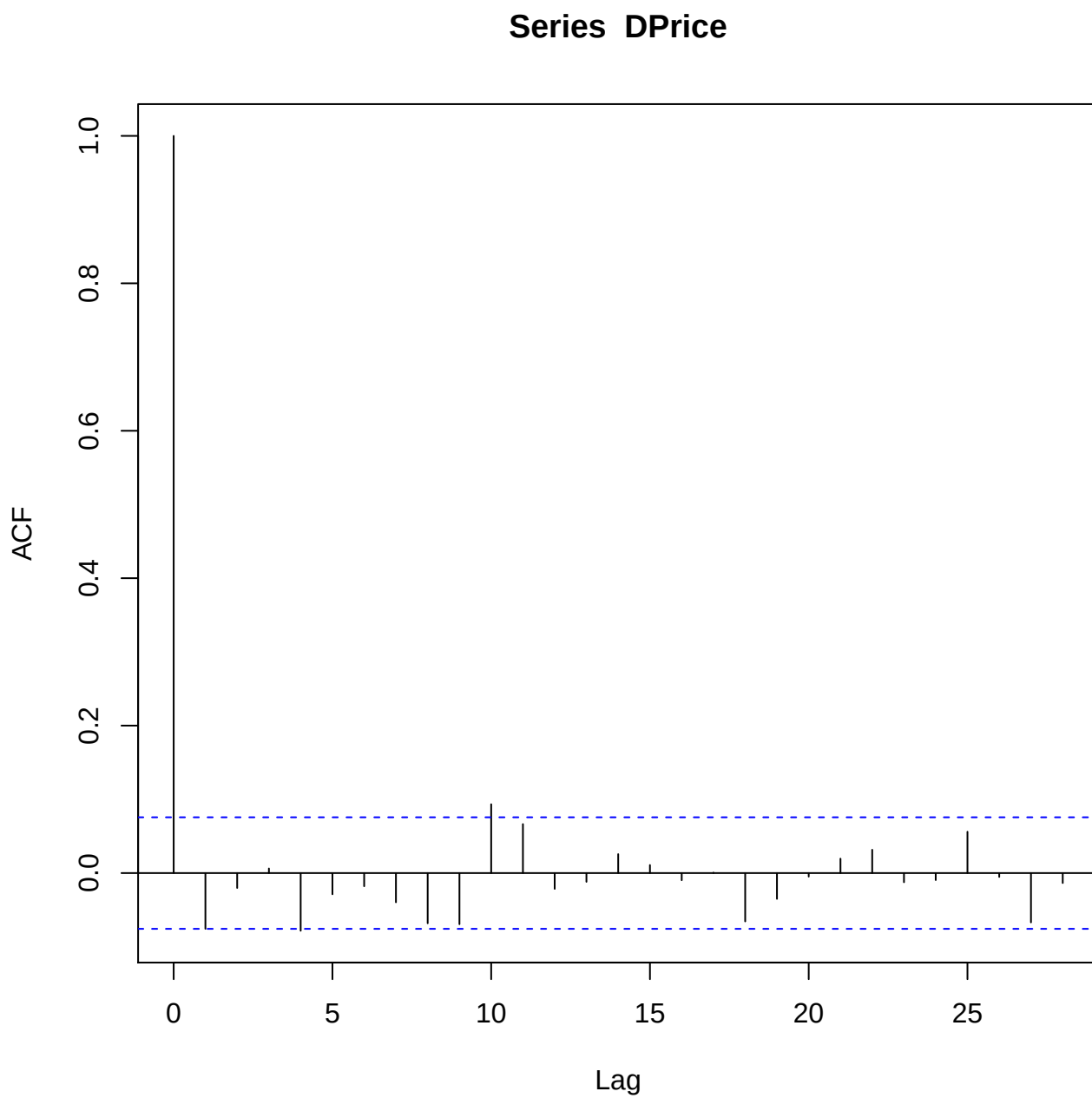


Figure 7: Correlogram of First Order Difference in HP Stock Prices



5 Autoregressive Processes

1. The time series $\{x_t\}$ is an *autoregressive process* of order p , abbreviated as $AR(p)$ if

$$x_t = \alpha_1 x_{t-1} + \alpha_2 x_{t-2} + \cdots + \alpha_p x_{t-p} + w_t,$$

where $\{w_t\}$ is white noise and the α_i are the model parameters with $\alpha_p \neq 0$.

2. Through an algebraic re-arrangement, we obtain

$$x_t - \alpha_1 x_{t-1} - \alpha_2 x_{t-2} - \cdots - \alpha_p x_{t-p} = w_t,$$

and using the backward shift operator notation B , we further obtain

$$\theta_p(B)x_t = (1 - \alpha_1 B - \alpha_2 B^2 - \cdots - \alpha_p B^p)x_t = w_t,$$

where $\theta_p(B)$ is a polynomial of B of order p .

3. The following points should be noted:

- The random walk is a special case of $AR(1)$ with $\alpha_1 = 1$ (see Section 3).
- The model is a regression of x_t on its past observations from the same series, hence the term “autoregressive”.
- The model parameters can be estimated by minimising the sum of square errors.

4. After estimating the parameters, we can predict the value of the series at time t and beyond provided we assume that the model parameters do not change in the prediction periods ahead. We use the formula

$$\hat{x}_t = \alpha_1 x_{t-1} + \alpha_2 x_{t-2} + \cdots + \alpha_p x_{t-p},$$

since the mean of w_t is zero.

5. The polynomial equation

$$\theta_p(B) = 0$$

is called the characteristic equation when B is treated formally like an algebraic variable. It does give us information about the time series if we solve for different values of B in the characteristic equation (the roots of the polynomial equation).

6. Since $\theta_p(B)$ is a polynomial of order p , it will have at most p roots, including repeated roots and/or complex roots when we solve $\theta_p(B) = 0$.
7. It turns out that all the roots B must satisfy $|B| > 1$ for the series to be stationary.
8. **Example 3.** The AR(1) model $x_t = \frac{1}{2}x_{t-1} + w_t$ is stationary because the characteristic equation is

$$1 - \frac{1}{2}B = 0,$$

and its only root is $B = 2$ which is greater than 1.

9. **Example 4.** The AR(2) model $x_t = x_{t-1} - \frac{1}{4}x_{t-2} + w_t$ is stationary. By using the backward shift operator notation, the model can be expressed as

$$(1 - B + \frac{1}{4}B^2)x_t = w_t.$$

Solving the characteristic equation

$$1 - B + \frac{1}{4}B^2 = 0$$

yields $\frac{1}{4}(B - 2)^2 = 0$ and thus we have repeated roots $B = 2$ which are both greater than 1.

10. **Example 5.** The random walk model $x_t = x_{t-1} + w_t$ in Section 3 is non-stationary. We rewrite the model as

$$(1 - B)x_t = w_t.$$

The characteristic equation is $1 - B = 0$ yielding the solution of $B = 1$ which is unity. Thus it is non-stationary and from our simulation example in Section 3, the plot of its path in Figure 4 shows that it is non-stationary.

11. When we have multiple roots, as long as the absolute magnitude of one of them is unity or strictly less than one, than the process is non-stationary.
12. For a complex root, such as $B = 0.3 + 0.4i$ where $i = \sqrt{-1}$, then its magnitude is calculated by $\sqrt{0.3^2 + 0.4^2} = 0.5$ which is less than 1 and thus such a series have this value of B as one of its root is non-stationary.
13. Also for complex roots, if $B = a + bi$ is a root, then we must have $B = a - bi$ as another complex root and the magnitude of either of them are $\sqrt{a^2 + b^2}$.
14. We will examine the autoregressive process in more detail in the next Chapter.
15. Other processes include the Moving Average process and the ARIMA model which combine both autoregressive processes and moving average processes. These will be examined separately.